

1. Explain Major Functionalities of Operating System.
2. What are different types of CPU Scheduling Algorithms? (FCFS, RR, PRI0, SJF)
3. Explain States of a Process in Operating Systems.
4. Virtual Memory in Operating System.
5. Explain Segmentation/Paging.
6. Can you explain what Hyper-V is?
7. What is Active Directory?
8. Explain basic features in Active Directory (AD) of Windows server?
9. Explain DNS & DHCP.
10. Explain What is VPN
11. Elaborate the concept of Network Policy Server (NPS)
12. What is difference between IPv4 and IPv6 addressing
13. What is File Server Resource Manager (FSRM) ?
14. What is IIS ? Which are the protocols that IIS supports?
15. Explain the concept of Network Load Balancing (NLB).
16. What is Powershell & What are the features of PowerShell?
17. Explain Linux Directory Structure in detail.
18. What is SSH & What is default port number of SSH ?
19. Explain each field of /etc/passwd
20. Why we use LVM in Linux & what are its benefits ?
21. How to create Filesystem ?
22. Explain Postfix Mail Server
23. What is postfix and default port used for postfix ?
25. What Is Dovecot?
26. What are FSMO roles in Windows Active Directory?
27. How to take active directory backup?
28. What is authoritative and non-authoritative restore of Active Directory?
29. How to transfer FSMO roles? What is seige FSMO role option? When to use it?
30. What is the difference between tree domain vs child domain?

COSA_Rapid_Fire_Sheet.txt

- What are built in groups - DomainAdmins , Enterprise Admins, Server Operator, Accounts Operator, Backup Operator, Domain Users
32. how to restore a failed Domain controller?
 33. How to recover forgotten Administrator password in Windows?
 34. How to recover forgotten root password in Linux?
 35. What is mounting in Linux?
 36. What is umask in Linux?
 37. What is SELinux?
 38. What is Nfs? What is root_squash?
 39. What is Network Information Service (NIS) in Linux?
 40. What is bashrc, .bashrc in Linux?

ITIM and DevOps

1. What is ITIL? What is V3?
2. Explain Software Development Life Cycle.
3. Explore Agile Development. What is scrum?
4. What is Data Center?
5. What is the functionality of a hypervisor?
6. What is Virtualization? What is difference between Type I, Type II virtualization, and containerization?
7. What is cloud computing? Explain cloud service models: IaaS, PaaS, SaaS, FaaS, DaaS.
8. What are the differences between vertical and horizontal scaling? Which one would you prefer and why?
9. How to create EC2 instance in AWS?
10. What is use of these AWS services - EC2, S3, SNS, EKS?
11. What is DevOps? Explain DevOps life cycle.
12. What is GIT? Explain GIT workflow.
13. What is GIT branching? How to implement it? Write commands to create branch, merge branch, and delete branch.
14. What is Docker? Explain Docker architecture.
15. What is Docker image and container? Explain life cycle of Docker container.
16. An application has two components i.e. MySQL database and Python Web application. Explain steps to containerize this application. How to use Docker compose?
17. What is orchestration? Which tools can be used for the same.
18. What is Kubernetes? Explain Kubernetes architecture.
19. What is pod? How do you scale pods?
20. What is CI? What is CD? Which are popular tools for CI/CD pipeline?
21. What is Jenkins? Explain Jenkins architecture.
22. How would you create pipeline in Jenkins?
23. What's the difference between continuous integration, continuous delivery, and continuous deployment?
24. What is Puppet? How it works?
25. What is Nagios? How it works?
26. Explain Service portfolio management.
27. Explain Business relationship management.
28. What is Service Design & its purpose?
29. What is Ansible, and how does it fit into the broader scope of configuration management?
30. Explain the key components of Ansible. What roles do these components play?
31. What are the benefits of using Ansible in a DevOps environment?
32. How does Ansible manage configurations on remote systems?
33. What is an Ansible playbook, and how is it structured?
34. What are handlers in Ansible playbooks, and when are they used?
35. What is an Ansible inventory, and what types of inventories can be used?
36. What is the purpose of inventory variables in Ansible, and how are they used?
37. What is an Ansible role, and how does it help with reusability?
38. What is Infrastructure as Code (IaC), and why is it important in modern IT operations?
39. What is Terraform, and how does it differ from other IaC tools like Ansible, CloudFormation, or ARM templates?
40. What are the key components of Terraform, and what roles do they play?
41. How do you initialize a Terraform working directory?
42. How do you manage sensitive data, such as API keys or passwords, in Terraform configurations?
43. How do you use variables in Terraform, and what are their benefits?
44. What are Terraform modules, and how do they enhance reusability?